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Effects of functional antioxidants on the expansion of gamma delta T-cells and their cellular cytotoxicity against bladder cancer cells

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Abstract

Purpose Results of previous studies have demonstrated that T-cell receptor cross-linking rapidly generates reactive oxygen species, which play essential signaling roles within mitochondria for the antigen-specific expansion of T-cells. However, oxidative stress also causes damage to cellular organelles. Thus, modulating ROS metabolism using antioxidants during naïve T-cell activation may promote the expansion and generation of functional T-cells. Notably, urothelial cancer is a sex-specific malignancy with high mortality rates worldwide. The present study aimed to evaluate the effects of various antioxidants on γδ T-cell proliferation, and the associated cytotoxicity against urothelial carcinoma cells (UCs).

Methods Over a period of cell induction and expansion, peripheral blood mononuclear cells were cultured with or without different antioxidants, including *N*-acetyl cysteine (NAC), vitamin C and vitamin E. Subsequently, phenotypic characterization of γδ T-cells and their cytolytic effects against UCs were analyzed by flow cytometry and cell viability assays, respectively.

Results and Conclusions The results revealed that NAC partially inhibited T-cell expansion in a dose-dependent manner. In addition, $CD3^+$/$V\gamma9^+$ levels and natural killer group 2D receptor expression were mildly reduced following treatment with a high dose of NAC, whereas $CD3^+$/$CD56^+$ levels and CD314 expression in natural killer-like cells were moderately decreased following treatment with vitamin E. Particularly, the direct co-incubation of bladder cancer cells with γδ T-cells supplemented with antioxidants significantly enhanced bladder cancer cytolysis. Collectively, results of the present study revealed that co-administration of functional antioxidants during γδ T-cell expansion may enhance the quality and efficacy of adoptive T-cell therapies for cancer treatment.

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Keywords Adoptive T-cell therapy, Γδ T-cell, Oxidative stress, Multi-functional antioxidant, Urothelial carcinoma of the bladder, Cytolysis

Introduction

Urothelial carcinoma (UC) of the bladder arises from the epithelial lining of the urogenital tract and is one of the most fatal malignancies worldwide [1]. Despite advancements in treatment, the overall survival rate of patients with late-stage UC has remained unchanged for the past two decades. In total, 25–30% of patients are initially diagnosed during the late stage of the disease, presenting with high-grade muscle-invasive bladder cancer (MIBC) [2]. Results of a previous study revealed that ~50% of patients with MIBC who undergo radical cystectomy may experience relapse, with a 5% survival rate observed in individuals with metastatic MIBC resistant to cisplatin [3]. Over the past decade, research has focused on immune checkpoint inhibitors that target programmed cell death protein 1 (PD-1) or programmed cell death protein ligand 1; however, results of these study revealed that a treatment response was only observed in 25% of patients. The inherent immunogenic variability of UC results in the expression of neoantigens, enabling the tumor to evade surveillance; thus, diminishing the anti-tumor response [4]. Therefore, the development of alternative treatments is required for patients with advanced MIBC.

γδ T-cells are a sub-population of unconventional T lymphocytes, accounting for <5% of the peripheral blood population. These cells function as innate T-cells that do not require T-cell receptor (TCR) engagement, functioning independently of major histocompatibility complex (MHC)-dependent antigens [5]. Thus, γδ T-cells exhibit potential in recognizing and killing tumors with high self-immunogenic variation, such as MIBC [6]. TCR activation triggers the Ras/Raf/MAPK-dependent oxidation pathway, resulting in the mass production of reactive oxygen species (ROS) within mitochondria and nicotinamide adenine dinucleotide phosphate (NADPH) oxidases [7]. Furthermore, elevation in ROS is one of the major characteristics of tumor microenvironment, which might lead to the infiltrating T cells experience oxidative stress and subsequently result in dysfunction and lack of persistence, and enhancement of antioxidant capability on T cells using thiol donors like N-acetylcysteine would rescue T cells from TCR stimulation-induced cell death and present superior anti-tumor capacity [8, 9]. Interestingly, it has also been reported that Vitamin E protected T cells from activation-induced cell death through inhibiting NF-kappa B activation and CD95 ligand expression [10]. Results of a previous study revealed that antioxidants improved T-cell memory and enhanced immunosurveillance in aged mice [11]. In addition, vitamin C significantly impacted T-cell differentiation and exerted direct effects on T-cell signaling [12]. In addition, vitamin D supplementation has been reported to increase the infiltrating CD8$^+$ T cells into tumors, with these cells exhibiting a more active (TEM/CM) of T cell phenotype [13]. More recently, a study demonstrated that *N*-Acetylcysteine synergized PD-1 antibodies to inhibit colorectal cancer progression in a mouse model, an effect mediated by CD8$^+$ T cells [14]. Results of our previous study revealed that adoptive T-cells, such as cytokine-induced killer T-cells or γδ T-cells, combined with chemoradiotherapy, may exert benefits against metastatic prostate cancer cells [15]. However, the effects of these multi-functional antioxidants on the proliferation and expansion of γδ T-cells, as well as the potential cytotoxicity against bladder cancer cells, have yet to be fully elucidated.

Thus, the present study aimed to evaluate the effects of various antioxidants on the expansion of γδ T-cells, and their cytotoxicity against UC cells. Over a two-week period of γδ T-cell induction and expansion, peripheral blood mononuclear cells (PBMCs) were cultured in the presence or absence of different antioxidants, including *N*-acetyl cysteine (NAC), vitamin C and vitamin E. Subsequently, the characteristics of γδ T-cells and the potential cytolytic effects against UC cells were assessed.

Materials and methods

Cell lines, reagents and chemicals

Human Grade III UC bladder cell line, T24 (cat. no. 60062), was purchased from the Bioresources Collection and Research Center. Recombinant human Proleukin® (interleukin-2) was purchased from Novartis International AG, and Zometa® (zoledronate) was obtained from Yungshin Pharm Ind. Co. Ltd. NAC, vitamin C, and vitamin E were purchased from Sigma Aldrich Inc. (St. Louis, MO, USA), and human peripheral blood mononuclear cells (hPBMC) were acquired from STEMCELL Technologies (Vancouver, BC, Canada). T24 cells were cultured in McCoy's 5a medium, purchased from Gibco; Thermo Fisher Scientific, Inc. Alamar Blue™ Cell Viability Reagent, GlutaMAX™ Supplement, non-essential amino acid solution (NEAA), sodium pyruvate, fetal bovine serum (FBS) and TrypLE™ Express Enzyme solution were also obtained from Gibco; Thermo Fisher Scientific, Inc. Human monoclonal antibody (mAb) isotype anti-IgG1, anti-CD3 mAb, anti-Vγ9 mAb, anti-CD56 mAb, anti-CD4, anti-CD8 mAb, anti-TCR mAb and anti-CD314 mAb were all purchased from BD Biosciences.

Culturing and storage of human bladder cancer cells

The T24 UC cell line was cultured in complete medium, consisting of 90% McCoy's 5a medium. This medium contained 1.5 g/l sodium bicarbonate, 10% FBS, sodium pyruvate, NEAA and glutamate (GlutaMAX™). Cells were cultured in a humidified cell culture incubator at 37 °C with 5% CO_2. In addition, cells were incubated until the 10th passage was reached, and subsequently stored in liquid nitrogen, using complete medium supplemented with 10% dimethyl sulfoxide (DMSO). TrypLE™ Express Enzyme solution was used for detaching the cells at each passage.

Maintain of PBMCs

PBMCs were thawed and resuspended in X-VIVO 15 basal medium (Lonza Group, Ltd., Lexington, MA, USA) supplemented with 5% human platelet lysate (EliteGro™, EliteCell Biomedical Corp.), and cultured in a humidified incubator at 37 °C with 5% CO_2. Amplified cells were preserved in commercial freezing medium for further experiments (CryoStor® CS10; Stemcell Technologies, Inc.).

Induction, expansion and antioxidant supplementation of γδ T-cells

The following methods were performed as previously described with modifications [16]. T-cells were induced on Day 0 in a G-Rex24 Well Plate (WILSONWOLF). Briefly, PBMCs were cultured in X-VIVO 15 basal medium supplemented with Proleukin® (IL-2, 1,000 U/ml), Zometa® (zoledronic acid, 5 μM) and human platelet lysate (hPL, 10%) for five consecutive days. From the fifth day onwards, the media was replaced every three days with fresh X-VIVO 15 media containing Proleukin® (2,000 U/ml). T-cells were harvested on the 14th day. Especially, the experimental group was cultured using the same method as the control group, with antioxidants added at different concentrations; namely, NAC (0, 1, 12 and 24 mM), vitamin C (0, 10, 100 and 200 μM), and vitamin E (0, 10, 100 and 200 μM). At the end of activation, cells were assessed for phenotypic characterization and cytolysis activity.

CD marker recognition for the assessment of γδ T-cells

After 14 days of culturing, T-cells were harvested and resuspended in PBS. Cell number and viability were assessed using the trypan blue dye exclusion assay with an automated cell counter (Cellometer Auto T4, Nexcelom, Waltham, MA, USA). Subsequently, the phenotype of T-cells was analyzed using flow cytometry. In total, >10,000 cells per specimen were used in the analysis. Briefly, T-cells were resuspended at 3×10^5 cells/100 μl PBS, and incubated for 30 min at 4 °C with the following anti-human monoclonal antibodies: Anti-CD3-pacificblue, anti-Vγ9, anti-CD56, anti-CD4, anti-CD8, anti-TCRαβ and anti-CD314. Corresponding isotype antibodies were used to stain negative control cells. Single cells were subsequently analyzed using a flow cytometer (SA3800 Spectral Analyzer; Sony Biotechnology).

Effects of antioxidant supplementation on the cytotoxicity of γδ T-cells against bladder cancer cells

In total, 5×10^3 T24 cells were seeded in a 96-well plate prior to the completion of γδ T-cell expansion. Subsequently, cell supernatants were replaced with 0.2 ml X-VIVO 15 complete medium with or without γδ T-cells as follows: i) Control group, 5×10^3 T24 cells (target cell, T) alone; and ii and iii) experimental groups, 2.5×10^4 γδ T-cells (effector cell, E) pre-induced with or without antioxidant were co-cultured with 5×10^3 T24 cells at the ratio of effector cells (E): target cells (T) = 5:1, respectively. The cell suspension was gently mixed, and the plate was returned to the incubator for an additional 24 h of incubation. Cell viability was assessed using the Alamar Blue assay. Briefly, cell supernatants were removed, and the adherent cells were washed twice. AlamarBlue™ reagent diluted in complete medium was added, and the plate was incubated for 3 h at 37 °C. Fluorescence (FL) was measured at 590 nm with an excitation wavelength of 545 nm using an ELISA reader (SpectraMax iD3, Molecular Devices, LLC). Cell viability was calculated as follows: (FL_treat - FL_control)/FL_control x 100.

Statistical analysis

Statistical analysis was conducted using SPSS software (version, 20.0; IBM Corp.). Results are presented as the mean ± standard deviation (SD), and each experiment was repeated at least three times. Group comparisons were performed using one-way ANOVA. $P<0.05$ was considered to indicate a statistically significant difference.

Results

Activation and expansion of T-cells from PBMCs with various antioxidants

The PBMCs were incubated with medium containing NAC (0–24 mM), vitamin C (0-200 μM) or vitamin E (0-200 μM). Subsequently, complete medium without zoledronic acid was refreshed on Days 5, 9 and 12, and cells were harvested on the 14th day (Fig. 1) due to the direct anti-proliferative and anti-metastatic effects of zoledronic acid against cancer cells [17]. Initially, 3×10^5 cells were seeded in each experimental group. Results of the present study demonstrated that the addition of NAC markedly inhibited the proliferation of T-cells in a concentration-dependent manner, with a partial ($P=0.074$) ~ 18.7% decrease in cell number observed at the highest concentration used. In addition, cell number was also decreased following treatment with the highest

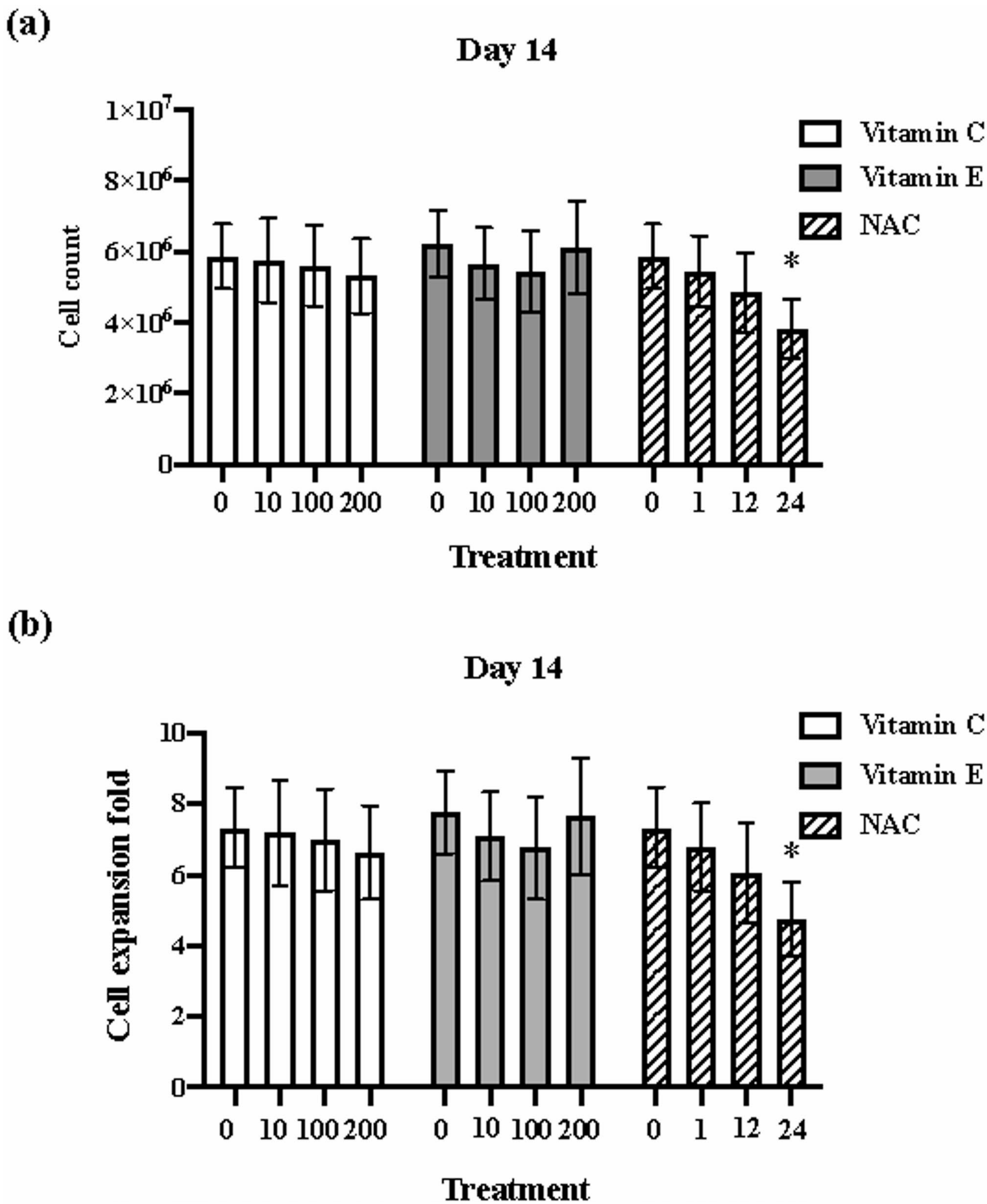

Fig. 1 Activation and expansion of γδ T-cells from PBMCs. The PBMCs were incubated with or without the medium containing NAC (0–24 mM), vitamin C (0-200 μM) or vitamin E (0-200 μM). Subsequently, complete medium without zoledronic acid was refreshed on Days 5, 9 and 12, and cells were harvested on the 14th day. The cell number was investigated by trypan blue dye exclusion assay. (**a**) Cell number per well of the G-Rex24 well plate. (**b**) Cell expansion fold following treatment and expansion. PBMCs, peripheral blood mononuclear cells. *, $P = 0.074$ compared with untreated group

dose of vitamin E, with a ~ 1.9% decrease in cell number observed.

Characterization of γδ/αβ T-cells

Following 14 days of culturing, γδ- and αβ-T-cells were analyzed. As shown in Fig. 2, the proportion of $CD3^+/V\gamma9^+$ cells was significantly increased following induction. In addition, the proportion of $CD3^+/V\gamma9^+$ cells was increased following treatment with vitamin E, while a slight decrease in the number of $CD3^+/V\gamma9^+$ cells was observed in vitamin C- and NAC-treated groups (Fig. 2a-g). Results of the present study also demonstrated that the proportion of $CD3^+/TCR\alpha\beta^+$ cells was markedly decreased following induction. However, no significant difference was observed between the control and antioxidant-treated groups, as shown in Fig. 2d and e.

Characterization of NK-like T-cells

Subsequently, natural killer (NK)-like T-cells were observed following γδ T-cell induction. As shown in Fig. 3a-c, the proportion of $CD3^-/CD56^+$ NK-like cells was significantly increased following induction. In addition, the proportion of $CD3^-/CD56^+$ NK-like cells was markedly increased following treatment with antioxidants; however, no change was observed following treatment with vitamin E. In addition, it also demonstrated that the number of $CD3^+/CD56^+$ NKT-cells was increased following treatment with antioxidants; however, treatment with high-dose vitamin E exerted a significant inhibitory effect on cell number, compared with the control group. Interestingly, it showed that the proportion of $CD314^+$ (NKG2D) T-cells was markedly increased following γδ T-cell induction; however, vitamin E and NAC exerted inhibitory effects (Fig. 3d-h).

Effects of antioxidant treatment on the cytolytic efficacy of γδ T-cells against urothelial cancer cells

In the present study, the potential cytotoxic effects of antioxidant-stimulated γδ T-cells against UC cells were investigated. It revealed that treatment with a 5:1 ratio

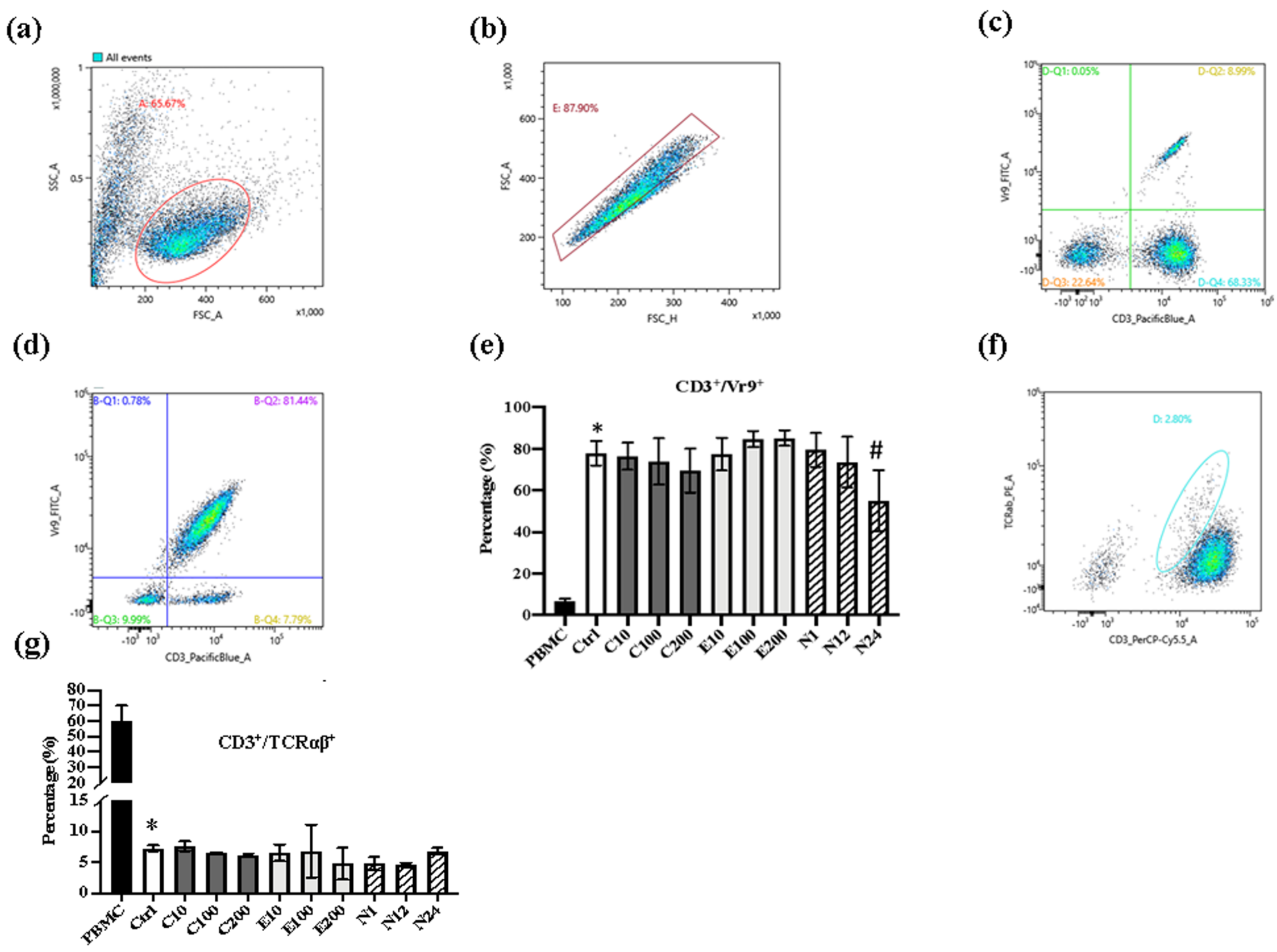

Fig. 2 Characterization of γδ/αβ T-cells. The antioxidant-induced γδ T-cells were investigated the expression of CD3, Vγ9, CD4, CD8, anti-TCRαβ for the characterization of γδ T-cells. (**a**) and (**b**), total cell illustration and the single cell gating strategy. (**e**, **f** and **g**) The proportion of $CD3^+/V\gamma9^+$ and $CD3^+/TCR\alpha\beta^+$ cells. (**c**) and (**d**) T cells before and after induction. *, $P < 0.05$ vs. PBMC group; $^\#$, $P < 0.05$ vs. control group. TCR, T-cell receptor

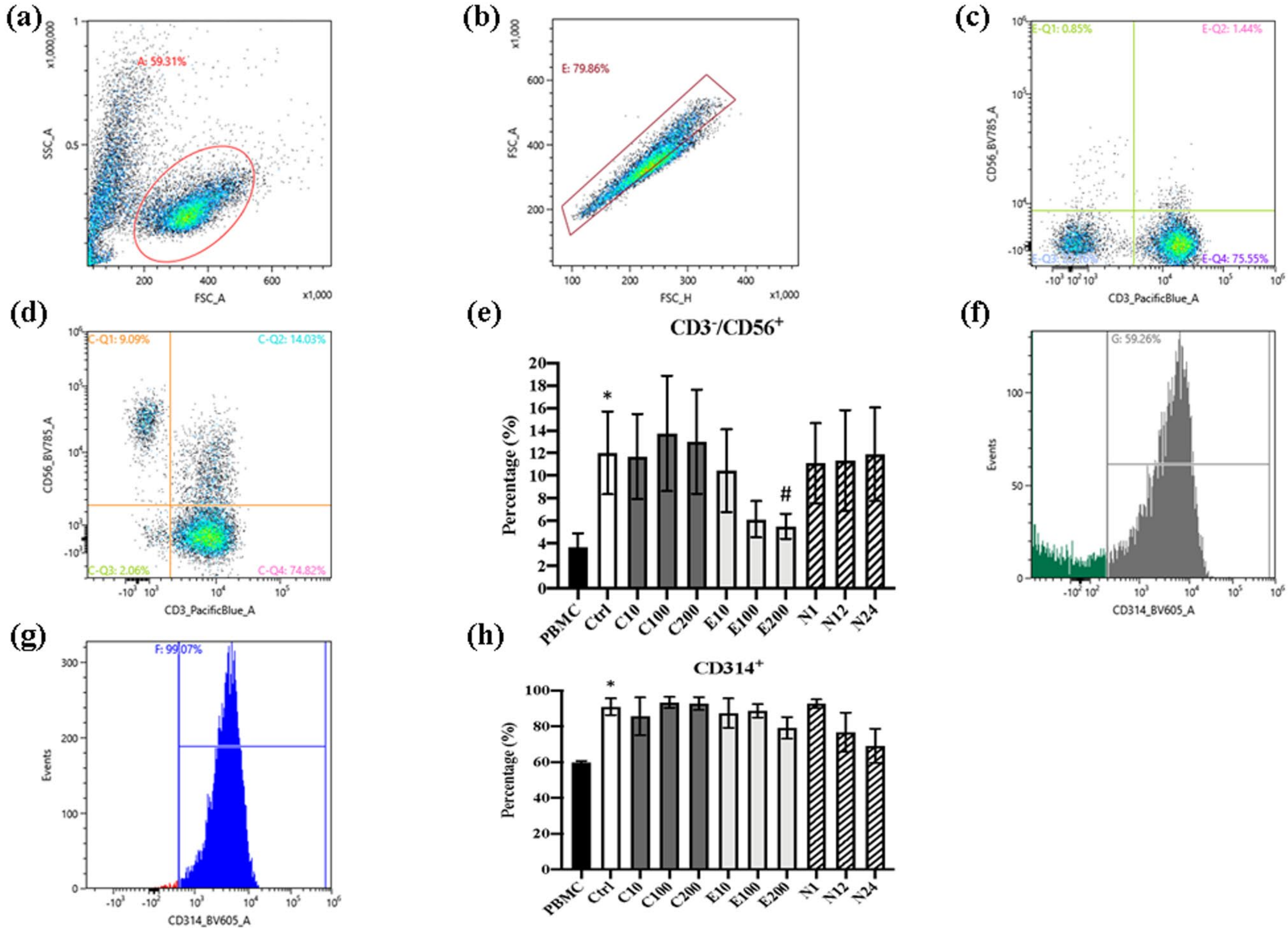

Fig. 3 Characterization of NK-like T-cells. The antioxidant-induced γδ T-cells were investigated the level of NK-like phenomenon. (**a**) and (**b**), total cell illustration and the single cell gating strategy. (**e** and **h**) The proportion of $CD3^{+}/CD56^{+}$ and $CD314^{+}$ cells. T cells before (**c** and **f**) and after (**d** and **g**) induction. *, $P < 0.05$ vs. PBMC group; $^{\#}$, $P < 0.05$ vs. control group. NK, natural killer cell; PBMC, peripheral blood mononuclear cell

of E: T cells induced significant cytotoxicity against UC cells following 24 h co-incubation (Fig. 4). In addition, we found that pre-stimulation of γδ T-cells with antioxidants increased the cytotoxicity of γδ T-cells against UC cells. Results also showed that the cytotoxicity of γδ T-cells against UC cells was most notable following treatment with the highest concentration of NAC.

Discussion

For numerous years, platinum-based chemotherapy has been considered the primary treatment option for patients with locally advanced or metastatic UC. Despite being the standard approach, the effectiveness of this therapy is limited, resulting in suboptimal treatment outcomes in patients and a low 5-year survival rate. Results of a previous study revealed that avelumab maintenance therapy may extend overall survival, compared with supportive care alone. However, numerous patients are diagnosed with late-stage disease, leading to limitations in potential treatment benefits. Results of a previous study revealed that a combination of nivolumab and gemcitabine-cisplatin may lead to a slight increase in the overall survival of patients. However, clinical trials that have explored the use of chemotherapy alongside immune checkpoint inhibitors have not demonstrated a significant improvement in overall survival outcomes for patients with locally advanced or metastatic UC [18]. Results of a previous study demonstrated that cytotoxic T lymphocytes (CTLs) play a key role in promoting immune responses against cancer through antigen-directed cytotoxicity [19]. In particular, T-cells use TCRs to directly interact with tumor antigens displayed on tumor cells via the MHC; thus, triggering the release of cytolytic mediators, such as granzyme and perforin. This leads to the amplification of tumor cell death [20]. γδ T-cells constitute a small subpopulation of human T lymphocytes, comprising 1–10% of all PBMCs. Furthermore, γδ T-cells express the γδ TCR [21]. Human Vγ9/Vδ2 T-cells, a subset of γδ T-cells, are particularly responsive to isopentenyl pyrophosphates, which are produced by various sources, including bacteria. The activation of these cells involves the interaction of butyrophilin with

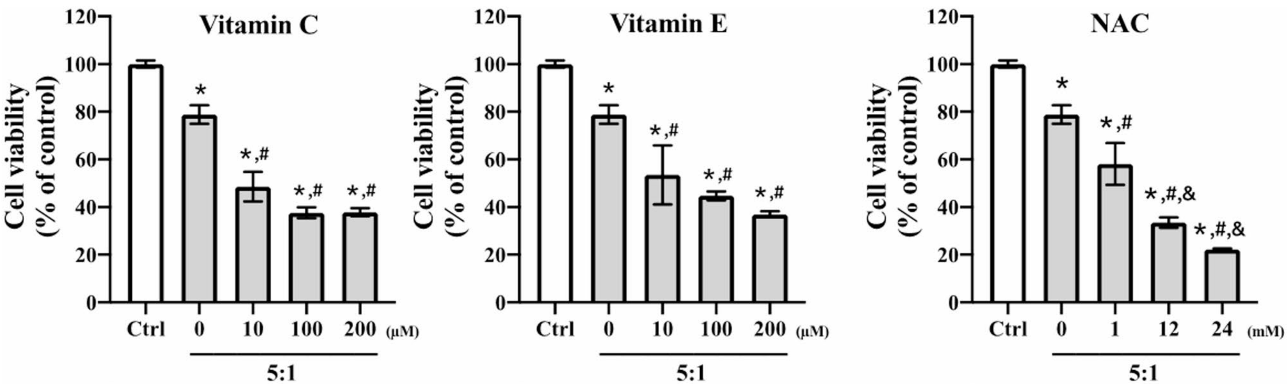

Fig. 4 Effects of antioxidant induction on the cytolytic efficacy of γδ T-cells against urothelial cancer cells. T24 cancer cells were co-cultured with the vitamin C- (**A**), vitamin E- (**B**) and NAC- (**C**) pre-induced γδ T-cells, respectively for 24 h. At the end of reaction, the suspended cells were removed and the viability of adherent cancer cells was evaluated by Alarma Blue assay. *, $P<0.05$ vs. control group; $^{\#}$, $P<0.05$, $^{\&}$, $P<0.01$ vs. γδ T group. NAC, *N*-acetyl cysteine

the corresponding phosphoantigen, leading to the subsequent binding to the non-variable region of the TCR. This mechanism renders γδ T-cells independent of human leukocyte antigen (HLA) molecules. However, when γδ T-cells are exposed to different stimuli during their development, they diversify into subpopulations with distinct characteristics. These subpopulations may exhibit either pro-tumorigenic or effector/memory polarization, distinguished by variations in surface markers, cytokine production and functional capabilities [21].

Improving the purity of γδ T-cells is crucial to address safety concerns regarding the potential mispairing of α/β chains with the endogenous α/β TCR or the reactivation of dormant self-reactive T-cells. In addition, this approach may mitigate reactivity against host molecules [21]. Notably, a Phase I clinical trial evaluated the safety and anti-tumor activity of genetically engineered autologous T-cells that express a specific TCR targeting UC [22]. In addition, results of a previous study revealed that activation of the Vγ9δ TCR promoted γδ T-cell cytotoxicity through increasing the secretion of perforin, granzymes and pro-inflammatory cytokines, such as IFN-γ and TNF-α. Besides, activation of the Vγ9δ TCR upregulated the expression of Fas ligand (FasL) and TNF-related apoptosis-inducing ligand (TRAIL), enabling γδ T-cells to target Fas^{+} and TRAIL-receptor^{+} tumor cells [23]. Furthermore, studies have shown that the glutathione pro-drug, *N*-acetylcysteine (NAC), regulates glucose and lipid metabolism in CD8^{+} T cells cultured in vitro and in vivo, promotes the conversion of CD8^{+} T cells to long-lived memory T cells (Tm), promotes TCF1 expression in CD8^{+} T cells, reduces their depletion and apoptosis, and thus enhances the anti-tumor ability of CD8^{+} chimeric antigen receptor T-cell transfusion (CAR-T) and lead to a reduce in CD8^{+} T cell exhaustion and induce their differentiation into long-lasting phenotypes, thus enhancing the anti-tumor effect of adoptive T cell transfer [14, 24–26]. These evidence indicated that NAC exerts pleiotropic effects that impact the quality of TCR-transduced T cells and recommend that the addition of NAC to current clinical protocols could be considered.

Results of the present study highlighted a significant increase in the proportion of CD3^{+}/Vr9^{+} cells and a decrease in CD3^{+}/TCRαβ^{+} cells following induction (Fig. 2). In addition, a lower efficiency of γδ T-cell induction was associated with a higher proportion of NK-like T-cells. Immune cells possess unique characteristics that allow them to sense environmental changes and activate appropriate responses, impacting their own functions, differentiation and proliferation [27]. Results of previous studies demonstrated that the characteristics and functional capabilities of T-cells are determined by the affinity of the TCR for the specific peptide-MHC complex on antigen-presenting cells and cytokines, and the concentrations of reactive oxygen and nitrogen intermediates [28]. An abundance of ROS may induce mitochondrial membrane depolarization, leading to inhibited T-cell activation and survival upon TCR engagement. Moreover, an excess of ROS may compromise the functionality of γδ T-cells and innate cellular responses to infection.

Recently, research focused on the impact of antioxidants on the activation of γδ T-cells and their cytotoxicity against cancer cells is limited. Results of a previous study highlighted that vitamin D3 may exert adverse effects on the expansion of γδ T-cells [29]. Moreover, the results also demonstrated that vitamin D3 suppressed IFN-γ production and negatively regulated the cytolytic effects of γδ T-cells against glioma cells [30]. Results of a previous study also indicated that vitamin C and its derivatives may enhance the proliferation of purified γδ T-cells and 14-day-expanded γδ T-cell lines [31]. In addition, treatment with NAC may promote γδ T-cell reconstitution, leading to the development of T-cells with a central memory phenotype [32]. Previously, an article revealed that NAC increases the antioxidant capacity of TCR-engineered T cells and improves in vivo tumor control and survival by improving antioxidant capacity, reducing DNA damage, and enhancing the ability of T cells to kill melanoma cells [25]. Furthermore, it is suggested that modulation of oxidative metabolism by antioxidants, in particular NAC, as one of feasible and inexpensive methods to generate $CD8^+$ stem-like memory T-cells [24]. Similar results were observed that NAC treatment would induced cytolysis of bladder cancer cells in dose-dependent manner; while it might have partial negative response on γδ T-cells proliferation when co-stimulation with high-dose NAC.

Research has also focused on increasing the effector functions and survival of primary $CD8^+$ CTLs and human CAR T-cells [33]. Notably, vitamin C enhanced the proliferation and effector functions of γδ T-cells; thus, leading to potential increases in the efficacy of γδ T-cells as a cancer immunotherapy using adoptive cell therapy [34]. Moreover, vitamin C plays a crucial role in facilitating the conversion of bone marrow progenitor cells into functional T-lymphocytes, and in the promotion of intrinsic T-cell development, a process that occurs independently of TCR rearrangement [35]. Results of previous studies demonstrated that pre-treating lymphoma cells or $CD8^+$ T-cells with vitamin C may increase T-cell-mediated cytotoxicity against lymphoma cells, enhance the infiltration of $CD8^+$ T-cells into tumors, and promote granzyme B production, when combined with anti-PD-1 treatment. Moreover, both vitamin C and its phosphorylated form may exhibit potential in promoting the effector functions of γδ T-cells; thus, promoting alternate anti-tumor effector cells, such as CAR T-cells or NK cells [36]. Much recently, the research revealed that STAT1 lysine vitcylation by vitamin C boosted anti-tumor immunity of $CD45^+$ $CD3^+$ T cells [37]. These results provided a mechanistic explanation for vitC's role in promoting anti-tumor-immune responses and showed promising therapeutic strategies combining vitC with immunotherapy. In conclusion, results of the present study revealed that co-stimulation of γδ T-cells with multi-functional antioxidants may enhance their cytolytic efficacy against UC cells. However, a limitation is that this is an in vitro experiment, even the use of patient-derived immune cells and further studies in animal models or humans are needed.

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Author contributions
Yueh Pan, Hung-Jen Shih, Pai-Fu Wang, Chi-Chen Lin, and Ping-Hsiao Shih contributed to the study conception and design. Yueh Pan, Shu-Han Chuang, and Ping-Hsiao Shih performed the experiments and data analysis. Yueh Pan, Hung-Jen Shih, Shu-Han Chuang, Chin-Pao Chang, and Chi-Hao Hsiao reviewed the literature and interpreted the results. Yueh Pan, Pai-Fu Wang, Chi-Chen Lin, and Ping-Hsiao Shih revised the manuscript. All authors read and approved the final manuscript.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Competing interests
The authors declare no competing interests.

Patient consent for publication
Not applicable.

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